

Math 118 – Lecture 04

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1 Main idea

Definition 1.1 (Orthogonal Complement). If V_0 is a subspace of V , the orthogonal complement $V_0^\perp$ of V_0 is

$$V_0^\perp = \{u \in V : \langle u, v \rangle = 0 \forall v \in V_0\}$$

Why is $V_0^\perp$ a subspace?

If

$$x, y \in V_0^\perp, \alpha \in K \leftarrow (\text{field of scalars: } \mathbb{R} \text{ or } \mathbb{C}) \text{ and } v \in V_0$$

Then, you have that

$$(\alpha x + y, v) = \alpha(x, v) + (y, v) = 0$$

because each of these terms is 0.

2 Definitions and notation

Definition 2.1 (First definition). Write the definition and one sentence explaining why it matters.

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example). Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: